

# Experiment 9

**Aim 1** : Copy a student object and modified the copied one without affecting the original one. (Clone of object)

## Code :

```
let student = { name:
"Aditya Wadhawan", age:
20, marks: 79
};
let clone = { ...student };

clone.marks = 95;

console.log(student); console.log(clone);
```

## OUTPUT :

Output:

19 ms | 44.5 MB

```
{ name: 'Aditya Wadhawan', age: 20, marks: 79 }
{ name: 'Aditya Wadhawan', age: 20, marks: 95 }
```

**Aim 2** : Create a student object where student has Name, Marks stored as object then remove one subject recalculate the total after removal of subject.

## Code :

```
let student = {
name: "Aditya",
marks: { math:
90, science:
85, english: 88
} };

```

```
delete student.marks.science;
let total = 0;
for (let sub in student.marks) { total
+= student.marks[sub];
}
console.log(student.marks);
console.log(total);
```

### **OUTPUT :**

```
Output: 18 ms | 45.0 MB
{ math: 90, english: 88 }
178
```

**Aim 3:** In the same previous ques, store Marks as subject wise object, find total of theory and practical subject wise for each student and then find total.

### **Code :**

```
let students = [
{ name:
"Aditya",
marks: {
math: { theory: 80, practical: 20 },
science: { theory: 70, practical: 25 }
}
}, { name:
"Krishna",
marks: {
math: { theory: 85, practical: 15 },
science: { theory: 75, practical: 20 }
}
}
];
```

```
let topTheory = { name: "", score: 0 }; let
topPractical = { name: "", score: 0 }; let
topOverall = { name: "", score: 0 };
```

```
students.forEach(s => { let
theory = 0, practical = 0;
```

```
for (let sub in s.marks) { theory +=
s.marks[sub].theory; practical +=
s.marks[sub].practical;
}
```

```
let total = theory + practical;
```

```
if (theory > topTheory.score) topTheory = { name:
s.name, score: theory };
if (practical > topPractical.score) topPractical = {
name: s.name, score: practical }; if (total >
topOverall.score) topOverall = { name: s.name, score:
total };
});
```

```
console.log(topTheory);
console.log(topPractical);
console.log(topOverall);
```

## **OUTPUT :**

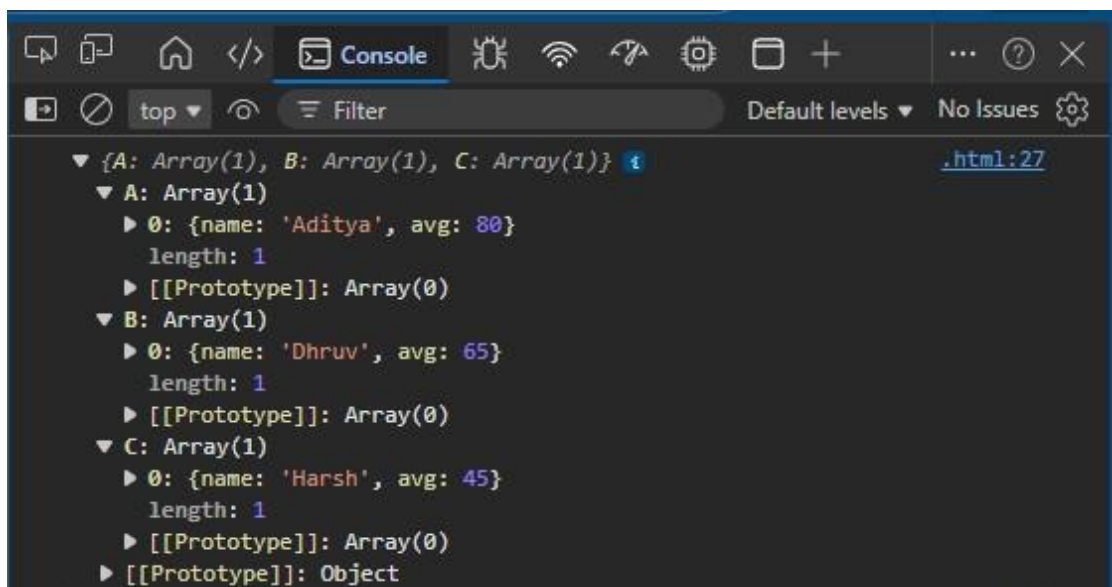
```
Output: 19 ms | 44.4 MB
{ name: 'Krishna', score: 160 }
{ name: 'Aditya', score: 45 }
{ name: 'Aditya', score: 195 }
```

**Aim 4:** Create array of students with Marks, Calculate average marks assigned grades (A,B and C) then group students into one object.

## Code :

```
let students = [  
  { name: "Aditya", marks: [80, 70, 90] },  
  { name: "Dhruv", marks: [60, 65, 70] },  
  { name: "Harsh", marks: [40, 50, 45] }  
];  
let result = { A: [], B: [], C: [] };  
  
students.forEach(s => {  
  let total = s.marks.reduce((a, b) => a + b, 0); let  
  avg = total / s.marks.length;  
  
  let grade = avg >= 75 ? "A" : avg >= 60 ? "B" : "C";  
  result[grade].push({ name: s.name, avg });  
});
```

## OUTPUT :

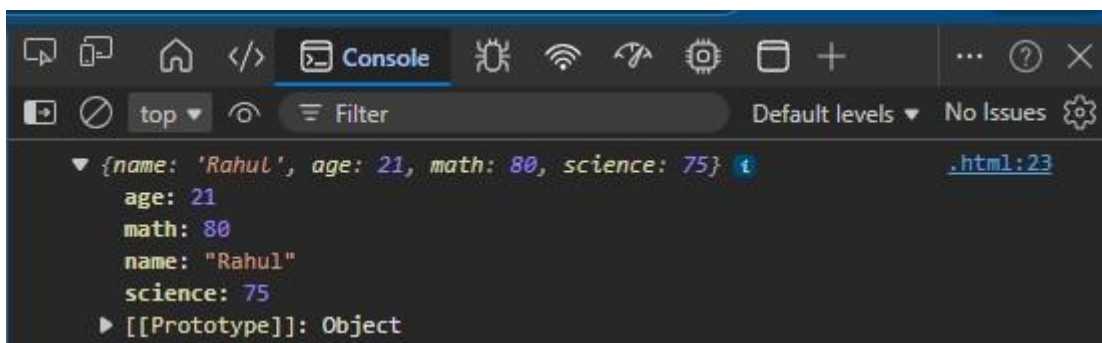


**Aim 5:** Create two objects, one contains student personal info like Name and Age other one contains Marks, Now merge both object into one, print the merged object.

### **Code :**

```
let personal = {  
  name: "Rahul", age:  
  21  
}; let marks  
= { math: 80,  
  science: 75  
};  
  
let merged = { ...personal, ...marks };  
console.log(merged);
```

### **OUTPUT :**



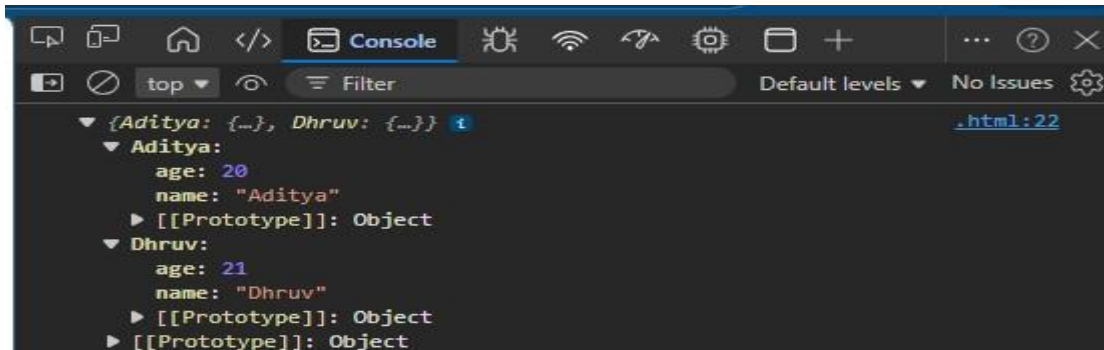
**Aim 6:** Given an array of students objects Stores name and age and now convert this array into object use student name as keys and store full student object as value

### **Code :**

```
let students = [  
  { name: "Aditya", age: 20 },  
  { name: "Dhruv", age: 21 }  
]; let obj =  
{};  
  
students.forEach(s => { obj[s.name]  
= s;  
});
```

```
console.log(obj);
```

## **OUTPUT :**



**Aim 7:** Create array of students where each student has Name, Marks in two tests and now calculate the total marks of test 1 and test 2 then find the student who needs highest improvement.

## **Code :**

```
let students = [
  { name: "Aditya", test1: 50, test2: 80 },
  { name: "Dhruv", test1: 60, test2: 70 },
  { name: "Gaurav", test1: 40, test2: 90 }
];
let maxImprovement = -Infinity; let
studentName = "";
students.forEach(s => {
  let improvement = s.test2 - s.test1;

  if (improvement > maxImprovement) {
    maxImprovement = improvement; studentName
    = s.name;
  } });
console.log(studentName, maxImprovement);
```

## **OUTPUT :**

